

MURUGESAN M

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PROFESSIONAL SUMMARY

Electronics and Communication Engineering undergraduate with strong foundation in Embedded Systems, IoT, Computer Vision and Machine Learning. Experienced in designing real-time embedded and AI-driven solutions using C/C++, Python and MATLAB. Demonstrated leadership, teamwork, communication and problem-solving through research projects, technical clubs and event management.

EDUCATION

Madras Institute of Technology, Anna University	Chennai
B.E. Electronics and Communication Engineering	2023–2027
CGPA: 8.16 (Up to 5th Semester)	
St. Antony's Matric Higher Secondary School	Tuticorin
HSC: 95.17%	May 2023

SKILLS

Programming: C, C++, Python, MATLAB

Embedded: Embedded C, ESP32, FPGA, Arduino, PCB Design

Tools: LTspice, Proteus, Altium Designer, CST Studio, Xilinx Vivado, Cadence, Cisco Packet Tracer, Git, Overleaf.

Core: Embedded Systems, IoT, Signal Processing, Machine Learning, Computer Vision, Communication Protocols, Hardware Design, Debugging

PROJECTS

Wrist Pulse Signal Classification for Vata, Pitta, Kapha Detection 2026

- Developing a wearable multi-sensor pulse acquisition system using Arduino and optical sensors for real-time physiological signal monitoring.
- Building a Python-based signal processing and machine learning pipeline for Dosha classification using multi-channel pulse waveform analysis.

IoT-Based Digital Letter Tracking and Management System 2026

- Developed an ESP32-based IoT document tracking system using Flask, PostgreSQL and QR-code technology for real-time workflow monitoring that reduced manual document processing by approximately 30%.
- Co-authored a research paper based on the project with secure cloud integration and automated document tracking.

Traffic Surveillance System 2025

- Developed a MATLAB-based traffic surveillance system using YOLOv8 and Kalman Filter for real-time vehicle detection, tracking and lane analysis.
- Implemented automated speed violation detection for intelligent traffic monitoring reducing manual monitoring by approximately 40% through automated vehicle detection.

Driver Drowsiness Detection System 2024

- Designed an embedded real-time driver monitoring system using IR sensors for eye-closure detection with sub-second alert response using IR sensors.
- Integrated alarm, LED indication and braking control to improve road safety.

PUBLICATIONS

Deployment of an IoT-enabled Letter Tracking System

- Published a co-authored research paper describing an ESP32-based IoT document tracking system integrating QR-code scanning, Flask backend and PostgreSQL cloud database for secure real-time document management.

LEADERSHIP & ACTIVITIES

Media Specialist | Event Organizer, QuantumIT Club Aug 2025

- Led social media campaigns, event promotions, logistics and public communication for Chakravyuha'25.
- Media Specialist and Digital Marketer at Araneri, contributing to branding, digital campaigns and website launches.

Volunteer | National Service Scheme (NSS) Present

- Participated in community outreach, awareness rallies and social initiatives.

EXPERIENCE

- Embedded Systems Workshop: Embedded C, Microcontrollers and Hardware Interfacing.
- Hands-on Workshop on RISC-V using FPGA and Xilinx Vivado.
- RUSA Workshop on Embedded Hardware Validation, Edge AI, RADAR Systems and PCB Fabrication at IIT Madras.

ACHIEVEMENTS

- Runner-Up, Q-Debate, Chakravyuha'25 (Classical vs Quantum Computing).
- Runner-Up, QUIZMOS, Astro Club, Anna University.
- Completed 5 km Swachh Bharat Cleanliness Awareness Marathon.

CERTIFICATIONS

- Python for Data Science – NPTEL
- MATLAB Onramp – MathWorks
- MATLAB Linear Algebra – MathWorks
- FPGA-Based Sensing for IoT – IEEE Webinar
- RISC-V on FPGA: Application and Porting – IEEE Student Branch MIT
- Quantum Computing and Quantum Simulation Workshop – QuantumIT Club, MIT